

The empirical recurrence for OEIS A300502, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A300502 records “Empirical recurrence of order 93” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 122$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 11$, so the recurrence is proved for every $n \geq 104$. Its last coefficient is $c_{93} = 4 \neq 0$, so no further tail satisfies anything shorter; any order-93 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A300502 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 1, 3 or 4 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

5, 3, 7, 20, 32, 68, 191, 504, ...

The entry states:

Empirical recurrence of order 93 (see link above)

As of the “Last modified” line on the live entry (Mar 07 2018 07:31:32, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 122$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 122$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 93: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 11$ is the first whose minimal order is exactly the stated 93.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 244$ exact terms from $n = 11$ on, it returns order 93 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{93} c_i a(n-i),$$

c_1	1	c_{25}	5753	c_{49}	382943	c_{73}	−7916
c_2	4	c_{26}	12848	c_{50}	−143514	c_{74}	2709
c_3	8	c_{27}	15436	c_{51}	−67113	c_{75}	17542
c_4	−2	c_{28}	−3885	c_{52}	20340	c_{76}	12021
c_5	−14	c_{29}	12115	c_{53}	−165130	c_{77}	4248
c_6	−22	c_{30}	−45143	c_{54}	−265111	c_{78}	−2099
c_7	−20	c_{31}	−25471	c_{55}	230147	c_{79}	−5391
c_8	−39	c_{32}	5885	c_{56}	12036	c_{80}	−3658
c_9	7	c_{33}	−59580	c_{57}	131489	c_{81}	−317
c_{10}	94	c_{34}	94765	c_{58}	291039	c_{82}	1052
c_{11}	186	c_{35}	42799	c_{59}	−56572	c_{83}	1351
c_{12}	15	c_{36}	40247	c_{60}	−100327	c_{84}	314
c_{13}	−180	c_{37}	42574	c_{61}	−51549	c_{85}	−376
c_{14}	−124	c_{38}	14963	c_{62}	−186311	c_{86}	−197
c_{15}	−216	c_{39}	−210636	c_{63}	−80682	c_{87}	−151
c_{16}	109	c_{40}	1438	c_{64}	87235	c_{88}	97
c_{17}	400	c_{41}	−18770	c_{65}	−1197	c_{89}	73
c_{18}	1455	c_{42}	−180475	c_{66}	83736	c_{90}	−8
c_{19}	960	c_{43}	312001	c_{67}	77851	c_{91}	−3
c_{20}	−146	c_{44}	−53138	c_{68}	−12414	c_{92}	−13
c_{21}	−3552	c_{45}	17115	c_{69}	9094	c_{93}	4
c_{22}	−4475	c_{46}	48188	c_{70}	−19156		
c_{23}	−2624	c_{47}	79814	c_{71}	−42586		
c_{24}	−3069	c_{48}	−272577	c_{72}	−18551		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 104$, and it is the recurrence OEIS A300502 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 11$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{93} - c_1x^{93-1} - \dots - c_{93}$. Its constant term is $-c_{93} = -4$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 93 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 93, so $\deg q = 93$, and it is monic. It holds from some index t on. From $\max(t, 11)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 93, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 27 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 93, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 4.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-93 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A300502.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.